

Pre- and postnatal toxicity of the commercial glyphosate formulation in Wistar rats

Eliane Dallegrave · Fabiana D. Mantese ·
Rosemari T. Oliveira · Anderson J. M. Andrade ·
Paulo R. Dalsenter · Augusto Langeloh

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Abstract Glyphosate is the active ingredient and polyoxyethyleneamine is the surfactant present in the herbicide Roundup® formulation commercialized in Brazil. The aim of this study was to assess the reproductive effects of glyphosate-Roundup® on male and female offspring of Wistar rats exposed during pregnancy and lactation. Dams were treated orally with water or 50, 150 or 450 mg/kg glyphosate during pregnancy (21–23 days) and lactation (21 days). These doses do not correspond to human exposure levels. The results showed that glyphosate-Roundup® did not induce maternal toxicity but induced adverse reproductive effects on male offspring rats: a decrease in sperm number per epididymis tail and in daily sperm production during adulthood, an increase in the percentage of abnormal sperms and a dose-related decrease in the serum testosterone level at puberty, and signs of individual spermatid degeneration during both periods. There was only a vaginal canal-opening

delay in the exposed female offspring. These findings suggest that in utero and lactational exposure to glyphosate-Roundup® may induce significant adverse effects on the reproductive system of male Wistar rats at puberty and during adulthood.

Keywords Male reproductive effects · Glyphosate · Polyoxyethyleneamine · Roundup® · Wistar rats

Introduction

Exposure of human populations to pesticides, industrial pollutants and synthetic chemicals present in foods, beverages and plastics has raised the concern that these substances could interfere with endogenous sex-hormone function. This interference can result in a variety of developmental and reproductive anomalies. Compounds in this class are thus referred to as endocrine disruptors (Akingbemi and Hardy 2001). Xenoestrogens could be implicated in the decreased male fertility and the increased incidence of testicular and breast cancers in humans. To predict their deleterious effects, various tests have been proposed to assay the activity of xenoestrogens (Massaad and Barouki 1999). In response to emerging concerns over the influences of endocrine disruptor chemicals the United States Environmental Protection Agency (US EPA) has established the Endocrine Disruptor Screening and Testing Advisory Committee (EDSTAC 1998). In its final report the EDSTAC proposed a Tier I screening battery to detect (anti)estrogenic(E), (anti)androgenic(A) and antithyroid activities using in vitro and in vivo assays. These tests have been used to detect if chemical substances or mixtures have potential to interact with the endocrine

E. Dallegrave (✉) · F. D. Mantese · A. Langeloh
Department of Pharmacology,
Federal University of Rio Grande do Sul,
Rua Sarmiento Leite 500 sala 202,
90046-900 Porto Alegre, RS, Brazil
e-mail: elianed@ufrgs.br

R. T. Oliveira
Departament of Pathology,
Federal University of Rio Grande do Sul,
Rua Bento Gonçalves 9090, 94500-000 Porto Alegre,
RS, Brazil

A. J. M. Andrade · P. R. Dalsenter
Department of Pharmacology,
Federal University of Paraná,
Jardim das Américas, 81531-990 Curitiba, PR, Brazil

system (Gray 1998). Moreover, in order to promote a broader screening approach additional protocols such as in utero and lactational studies are being considered as alternative screening assays (EDSTAC 1998).

Pesticides are economically important chemicals in agriculture, and their use has increased the yield of plants for food resulting in reduced food costs. However, they generally persist in the agricultural products and in the environment, posing health hazards to humans and animals (Dalsenter et al. 1999). Despite many pesticides being less persistently used, others are still moderately or persistently used, principally in the developing countries.

Sexual development during the prenatal and neonatal periods is under hormonal control and is therefore sensitive to exogenous substances with endocrine effects. Disturbances of hormonal regulation during prenatal or early postnatal development may induce adverse effects on the male reproductive system (Mably et al. 1992). The endocrine modulation potential of glyphosate has been evaluated in a variety of studies, including in vitro assays and standard in vivo toxicology studies (Williams et al. 2000). The reproductive toxicity of glyphosate was studied in rats, and the reported reduced epididymal sperm concentrations (20% below control) were at 25,000 and 50,000 ppm levels; nevertheless, all values were well within the normal range of their historical control data (Morrissey et al. 1988). And also, the reproductive toxicity of glyphosate was studied in rabbits, and the reported male toxic effects included reduced ejaculate volume and sperm concentration, and increased abnormal and dead sperm (Yousef et al. 1995). But these results, according to Williams et al. (2000), could not be properly interpreted, because the small group-size, young age of its members and other protocol data failed to support the conclusions. Based on these limitations of the study on rabbits as well as considering the above results with rats, these reproductive results cannot be used to support the possibility of the glyphosate to disrupt the endocrine system. However, Walsh et al. (2000) demonstrated that Roundup (commercial formulation) inhibited steroidogenesis by disrupting expression of the steroidogenic acute regulatory (StAR) protein, showing that this commercial formulation of the pesticide has the potential to disrupt reproductive function in animals. Richard et al. (2005) showed that the glyphosate-Roundup® decreased the aromatase enzyme activity on human placental cells through in vitro assays on smaller concentrations as used in agriculture. This enzyme is responsible for the estrogen synthesis. Low levels of estrogens in the brain during critical periods like sex differentiation for the male can affect the fertility at puberty or adult age (Kelly 1991).

The Roundup® formulation, which contains glyphosate as the active ingredient and polyoxyethyleneamine (POEA) as the surfactant agent, is marketed as a non-selective, broad-spectrum, post-emergence herbicide used to control weeds in emerged grasses, broad-leaf weeds, pastures and cultures such as rice, corn and soy (Menkes et al. 1991; Smith and Oehme 1992; WHO 1994). The proportion of glyphosate to surfactant in the Roundup® formulation varies according to the country in which the product is marketed (Martinez et al. 1990). In Brazil, Roundup® is available as 36% glyphosate and 18% polyoxyethyleneamine. Dallegrave et al. (2003) demonstrated that the Roundup® formulation, commercialized in Brazil, is toxic to dams treated with 1,000 mg/kg of the glyphosate (of this formulation) from day 6 to 15 of pregnancy (with 50% of lethality), considered as NOEL-dose of the glyphosate alone for this period (WHO 1994). The oral acute toxicity of this formulation was studied by Dallegrave et al. (2002) and also showed more toxicity ($LD_{50} = 2.3$ g/kg) than reported in the literature (WHO 1994) for glyphosate alone ($LD_{50} > 5$ g/kg).

Considering the possibility of this formulation to be more toxic than glyphosate alone and/or disrupt the endocrine system, the aim of the present study was to elucidate whether glyphosate-Roundup® (commercial formulation) poses reproductive hazards to male and female offspring of rats exposed during pregnancy and lactation.

Materials and methods

Animals

Sexually mature male and female Wistar rats (90 days old, 250–350 g) bred in the animal facilities of our own department were used. All breeding phases and all experiments were performed in accordance with the rules of the Ethics and Animal Experimentation Committee. All animals were housed in polyethylene (65 × 25 × 15 cm) home cages with sawdust-covered floors. Animals were maintained in a colony room at $22 \pm 2^\circ\text{C}$ and on a 12 h-light/dark cycle (lights on from 9 a.m. to 9 p.m.) with free access to standard lab rat chow (Nuvital®, Curitiba/PR, Brazil) and water.

Chemicals

The Roundup® formulation (Monsanto of Brazil) contained 360 g/l of glyphosate (N-phosphonomethylglycine) and 18% (w/v) polyoxyethyleneamine (surfactant). The solutions of the Roundup® formulation were

prepared by the addition of appropriate volumes of distilled water.

Mating

Three female rats were placed in a cage with one male during the dark period (between 9 p.m. and 9 a.m.) for mating. On the subsequent morning (9 a.m.), vaginal smears were obtained from all females and examined. Females showing sperm were housed in individual cages, and this was considered the zero day of pregnancy (Paumgarten et al. 1997). The other females were returned to the cage of the same male, each dark period during 15 consecutive days (mating period).

Treatment

Sixty primigravid female rats were randomly divided into 4 groups of 15 animals each and treated orally by gavage (a volume of 10 ml/kg) with 0, 50, 150 or 450 mg/kg glyphosate-Roundup® during pregnancy (21–23 days) and lactation (21 days).

Maternal and reproductive outcome data

Body weight was recorded daily during the pregnancy and lactation periods. The body weight noted at day 0 (first period day) was considered as 100%, for each period. The differences observed during the experiment with regard to this parameter were expressed as relative weight gain. Litter size, number of living and dead pups, viable pups, and sex ratio (male/female pups) were recorded.

Offspring development

The pup variables included weekly evaluation of body weight (g) from lactation to puberty and daily evaluation of the development of general and sexual characteristics until puberty. The characteristics included ears unstuck, fur emergence, incisor eruption, eye opening, testis descent (by scrotum palpation starting after the 15th postnatal day), preputial separation (by manually retracting the prepuce with gentle pressure after the 30th postnatal day) and opening of the vaginal canal (after the 30th postnatal day).

Males

Euthanasia

One male from each litter was randomly ($n = 15/\text{group}$) selected to investigate the systemic and repro-

ductive effects of treatment at puberty (65 days of age). The animals were euthanized from 8 a.m. to 9 a.m. after being anesthetized with 50 mg/kg thiopental injected intraperitoneally. The abdomen was then incised the vena cava was exposed and blood was collected. Under anesthesia, the diaphragm was incised to kill the animal. One male from each litter was randomly ($n = 15/\text{group}$) selected to investigate the systemic and reproductive effects of treatment during adulthood (140 days of age). The animal was euthanized by the same procedure as described for puberty and systemic toxicity was determined on the basis of relative organ weight expressed as percentage of body weight. Male reproductive toxicity was determined on the basis of relative weight of the reproductive organs expressed as percentage of body weight and of reproductive indices, including sperm number per epididymis tail, daily sperm production, sperm transit, sperm morphology, testis morphology and serum testosterone level (US EPA 1996).

Systemic organs

Heart, lungs, liver, spleen, kidneys, adrenal glands and brain were removed and weighed. The organ weight was related to body weight. This parameter was expressed as relative systemic organ weight.

Sex organs

The testis, epididymis, seminal vesicle with coagulating glands (without fluid) and prostate were removed and weighed. The organ weight was related to body weight. This parameter was expressed as relative sex organ weight.

Spermatid and sperm count

For spermatid number count, after removal of the tunica albuginea, each testis was minced and homogenized in 10 ml of 0.9% NaCl containing 0.5% Triton X-100 at medium speed in a FISATOM 720 tissueizer® for 1 min. After dilution in 0.9% NaCl, the number of homogenization-resistant spermatids of each testis was counted with a hemocytometer. The number of spermatids per animal (right plus left testis) was divided by 6.1 days to convert the value to daily sperm production (Robb et al. 1978). Similarly, the epididymis tail was cut into small pieces, minced and homogenized, and the spermatozoa were counted as described above. The epididymal sperm transit rate was calculated by dividing the epididymal sperm number by daily sperm production (Amann et al. 1976).

Sperm morphology

To assess the percentage of morphologically abnormal sperm (detected in the head or tail piece) the deferens ducts were rinsed with 0.5 ml 0.9% NaCl (for 65-day-old animals) or 1 ml 0.9% NaCl (for 140 day-old-animals), and a sperm suspension was obtained. An aliquot of sperm suspension was carefully stained with 2% eosin to prepare a smear on the slide. Two hundred sperm per animal were analyzed microscopically at 400× magnification and morphologically normal and abnormal sperm were recorded according to the presence or absence of defects found in the head or tail of the spermatozoon (Robb et al. 1978).

Histology

For the histological study, five testes per group were fixed in Bouin's solution immediately after removal, embedded in paraffin, sectioned at 3 µm and subsequently stained with hematoxylin/eosin (Russell et al. 1990). Twenty essentially round seminiferous tubules per testis were analyzed microscopically at 100× magnification to assess the mean tubule diameter. Moreover, 100 tubules per testis were analyzed microscopically at 400× magnification to assess the percentage of seminiferous tubules with complete spermatogenesis (tubules with elongated spermatids). The presence of degenerating, sloughed and/or infiltrating cells and the absence of tubular lumen and of elongated spermatids were analyzed microscopically at 400× magnification.

Testosterone level

For testosterone level determination, blood was collected and serum removed by centrifugation and stored at –20°C until analysis. The samples were analyzed in duplicate using a double-antibody according to the standard protocol for radioimmunoassay (RIA) with Diagnostic Products Corporation (Los Angeles) testosterone kits. The mean result was reported in ng/ml and the intra- and interassay variations were less than 10%.

Females

Euthanasia

One female from each litter was randomly selected ($n = 15/\text{group}$) to investigate the systemic and reproductive effects at puberty (65–70 days old). On the first estrus after 65 days of age, the rats were anesthetized intraperitoneally with 50 mg/kg thiopental. Under

ether anesthesia, the diaphragm was incised to kill the animal. One female from each litter was randomly selected ($n = 15/\text{group}$) to investigate the systemic and reproductive effects during adulthood (140 days old), on the first estrus after this period. The protocol was the same as described for puberty. Female systemic toxicity was evaluated by relative weight of the organs (expressed as percentage of body weight) after euthanasia. Female reproductive toxicity was evaluated by relative weight of the reproductive organs (expressed as percentage of body weight).

Systemic organs

Heart, lungs, liver, spleen, kidneys, adrenal glands and brain were removed and weighed. The organ weight was related to body weight. This parameter was expressed as relative systemic organ weight.

Sex organs

The uterus, oviducts and ovaries were removed and weighed. The organ weight was related to body weight. This parameter was expressed as relative sex organ weight.

Statistical analysis

Parametric data, expressed as mean $\pm$ standard error (SEM), were analyzed by repeated measure ANOVA or one-way ANOVA, followed by the Bonferroni test when appropriate. The nonparametric data, expressed as proportion or percentage, were analyzed by the chi-square test. Differences were considered statistically significant when $P < 0.05$.

Results

Maternal and reproductive outcome data

No sign of maternal toxicity such as mortality or a significant decrease in the relative body weight gain of the dams during pregnancy and lactation was observed. The offspring variables, including litter size, number of living and dead pups, viable pups, and sex ratio (male/female pups) were also unaffected (Table 1).

Offspring development

Pup body weight (g) during lactation and up to puberty and the development of general characteristics did not show significant differences between genders. However,

Table 1 Maternal and reproductive outcome of dams exposed to glyphosate-Roundup[®] throughout pregnancy and lactation

Parameter	Glyphosate-Roundup [®] (mg/kg)			
	0	50	150	450
Relative weight gain (%) during pregnancy ^{a,b}	36.8 ± 2.1	38.7 ± 1.5	38.8 ± 1.9	40.9 ± 1.6
Litters (pups)	15 (160)	15(161)	15(165)	15(162)
Sex ratio (males/females)	75/85	80/81	83/82	93/69
Live birth index (%) ^c	99.4	99.4	99.4	100
Weaning index (%) ^d	95.0	98.8	97.0	90.7
Litter size (n) ^a	10.7 ± 0.69	10.8 ± 0.68	10.3 ± 0.77	10.8 ± 0.66
Birth weight (g) ^a	5.77 ± 0.04	5.97 ± 0.04	5.90 ± 0.04	6.05 ± 0.04
Weaning weight (g) ^a	31.6 ± 0.69	32.2 ± 0.45	32.6 ± 0.47	31.7 ± 0.48

^a Mean ± SEM^b Relative weight gain = relative weight on the last day of pregnancy minus relative weight on the first day of pregnancy (weight on the first day of pregnancy = 100%)^c Live birth index = (no. of live offspring/no. of offspring delivered) × 100^d Weaning index = (no. of live pups on day 21/no. of live offspring born) × 100

the male sexual characteristics showed a tendency to advance, while the female sexual characteristics showed a delay. The vaginal canal opening was significantly delayed for all dose levels. The preputial separation was significantly advanced for the high dose-level, but the testis descent did not show a significant difference ($P = 0.057$; ANOVA; (Table 2). These differences were statistically significant but did not show biologic significance. The sexual characteristics for both genders were manifested within the physiological period for this species (Baker et al. 1980). And also, all values were well within the normal period of the historical control data for these Wistar rats in our own department.

Male outcome data

Male sex organs did not show significant differences (Table 3). However, the percentage of abnormal sperm increased significantly at puberty, and daily sperm production and sperm number decreased significantly

during adulthood (Table 4). The testosterone concentration decreased dose-relatedly at puberty ($P = 0.051$, ANOVA) and the percentage of abnormal sperm tended to increase during adulthood ($P = 0.066$, ANOVA) at all dose levels (Table 4).

Histology of the testis showed the presence of growth disorders and degeneration characterized by a decrease in elongated spermatid and the presence of vacuolization at puberty for the medium (in four out of five testis) and high dose levels (in four out of five testis; Fig. 1a). Histology of the adult testis showed the presence of intense degeneration characterized by the absence of tubular lumen for all doses (low in three out of five, medium in four out of five and high in four out of five testis) (Fig. 1b).

Female outcome data

Neither the systemic nor reproductive organs of females were deeply affected at puberty or during

Table 2 Effects of in utero and lactational glyphosate-Roundup[®] exposure on body weight (g) and sexual characteristics

Parameter	Glyphosate-Roundup [®] (mg/kg)			
	0	50	150	450
Number of male pups	71	88	83	85
Day of testis descent ^a	15.0 ± 0.04	15.0 ± 0.05	15.0 ± 0.06	14.8 ± 0.06
Body weight at testis descent (g)	20.4 ± 0.38	21.0 ± 0.22	21.1 ± 0.27	20.3 ± 0.28
Day of preputial separation ^a	31.7 ± 0.29	31.7 ± 0.24	31.5 ± 0.24	30.7 ± 0.12*
Body weight at preputial separation (g)	73.0 ± 1.09	68.1 ± 0.60	72.2 ± 1.01	70.7 ± 0.80
Number of female pups	81	72	81	61
Day at opening of the vaginal canal ^a	34.9 ± 0.33	37.6 ± 0.40*	36.9 ± 0.47*	36.7 ± 0.46*
Body weight at opening of the vaginal canal (g)	75.1 ± 1.26	68.8 ± 1.32	72.3 ± 1.11	72.3 ± 0.93

^a Mean ± SEM*Significantly different from the control group: $P < 0.05$; ANOVA—Bonferroni test

Table 3 Effects of in utero and lactational glyphosate-Roundup[®] exposure on relative (%) sex organ weights of male offspring rats at puberty and during adulthood

Age (days)	65				140			
Glyphosate-Roundup [®] (mg/kg)	0	50	150	450	0	50	150	450
Testis (%) ^a	1.13 ± 0.04	1.04 ± 0.03	1.11 ± 0.02	1.13 ± 0.06	0.76 ± 0.02	0.81 ± 0.02	0.81 ± 0.02	0.74 ± 0.02
Epididymis (%) ^a	0.27 ± 0.009	0.26 ± 0.005	0.25 ± 0.007	0.28 ± 0.013	0.33 ± 0.009	0.33 ± 0.009	0.34 ± 0.008	0.31 ± 0.010
Seminal vesicle (%) ^a	0.17 ± 0.010	0.15 ± 0.007	0.15 ± 0.007	0.18 ± 0.014	0.20 ± 0.008	0.22 ± 0.014	0.21 ± 0.012	0.23 ± 0.011
Prostate (%) ^a	0.09 ± 0.005	0.07 ± 0.004	0.10 ± 0.004	0.10 ± 0.006	0.13 ± 0.008	0.13 ± 0.008	0.14 ± 0.008	0.15 ± 0.011

^a Mean ± SEM**Table 4** Effects of in utero and lactational glyphosate-Roundup[®] exposure on sperm number in the cauda epididymis, daily sperm production, sperm transit rate, testosterone level, diameter of seminiferous tubules, percentage of seminiferous tubules with complete spermatogenesis and sperm morphology

Age (days)	65				140			
Glyphosate-Roundup [®] (mg/kg)	0	50	150	450	0	50	150	450
Daily sperm production (x 10 ⁶) ^{a,b}	11.1 ± 0.7	12.2 ± 0.7	12.1 ± 1.0	11.7 ± 0.5	20.5 ± 1.7	15.3 ± 1.1*	19.7 ± 1.9	14.7 ± 1.0*
Sperm number (x 10 ⁶) ^{a,b}	44.2 ± 4.2	53.9 ± 7.1	67.2 ± 11.2	57.4 ± 6.7	344.7 ± 30.8	251.0 ± 18.7*	368.7 ± 36.0	257.1 ± 17.9*
Sperm transit rate (days) ^{a,b}	4.1 ± 0.4	4.6 ± 0.7	5.3 ± 0.6	5.1 ± 0.7	17.7 ± 1.5	17.5 ± 1.8	20.2 ± 2.2	18.5 ± 2.0
Abnormal sperm (%) ^b	8.6 ± 0.9	16.7 ± 1.9*	9.2 ± 0.8	11.6 ± 1.3	5.4 ± 0.5	8.3 ± 1.1	8.4 ± 1.0	7.7 ± 0.7
Tubules with spermatogenesis (%) ^c	84.0 ± 3.4	77.0 ± 2.9	78.7 ± 3.5	75.0 ± 2.1	92.0 ± 1.2	73.5 ± 2.8	74.5 ± 1.9	65.0 ± 3.7
Tubule diameter (μm) ^{a,c}	166.7 ± 0.4	159.6 ± 0.4	159.6 ± 0.7	160.4 ± 0.1	180.8 ± 0.3	187.3 ± 1.2	173.4 ± 0.3	185.1 ± 0.4
Testosterone level (ng/mL) ^{a,b}	5.2 ± 1.2	4.0 ± 1.0	3.2 ± 1.1	1.5 ± 0.2*	3.9 ± 0.8	3.4 ± 0.4	6.3 ± 1.0	3.3 ± 1.3

^a Mean ± SEM^b N = 15^c N = 5*Significantly different from the control group: *P* < 0.05; ANOVA—Bonferroni test

adulthood by pre- and postnatal exposure to glyphosate-Roundup[®]. Only the vaginal canal opening was delayed for all dose levels.

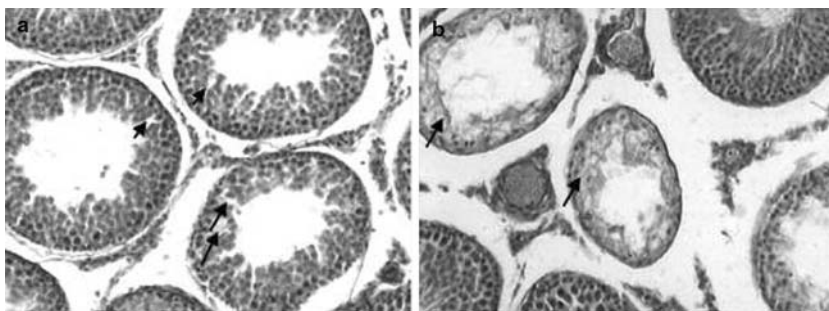
Discussion

In this study, male sexual characteristics like mean age at testis descent or preputial separation did not show differences from physiological responses (Baker et al. 1980). These variables are external signal of sexual development in male rats and are androgen secretion dependent (Waller et al. 1995). The relative weights of

male sexual organs were not affected; nevertheless, the lack of effect on organ weight should not be used to overpower significant changes in other endpoints that may be more sensitive (Dalsenter et al. 1999). The best male reproductive outcomes to be evaluated in toxicity studies are the testis relative weight, testis histology, sperm number and morphology (Amann 1982).

Our results showed that there were significant differences in testosterone levels in treated animals. This variable showed a dose-related decrease at puberty. Reproductive development depends on endocrine communication throughout the mammalian life cycle. Protein and steroid hormones, growth factors and

Fig. 1 Testes histopathological analysis of the rats exposed to the glyphosate-Roundup® during pre- and postnatal periods: **a** elongated spermatid vacuolization in the testes at puberty, **b** tubular degeneration in the testes at adult period



other signaling molecules affect gene expression and protein synthesis in target cells of different tissues. It is likely that the developing fetus is more sensitive to the effects of environmental pollutants than the adult organism; however, these in utero effects may not be manifested until puberty or adulthood (WHO 2001). In the present study the male toxicity characterized by reduced sperm counts in epididymis tail and decreased daily sperm production during adulthood might have resulted from chemical interference of the herbicide glyphosate-Roundup with the normal hormonal balance and normal functioning of important reproductive components such as Sertoli and/or Leydig cells in the developing male.

Similarly, the increased percentage of abnormal sperms at puberty and the results of testis histology that showed spermatid vacuolization at puberty and degeneration and absence of tubular lumen during adulthood may have resulted from changes in testosterone levels at puberty that showed a dose-related decrease. Normally, most male sex organs are fully imprinted within 10 to 20 days after birth (Amann 1982). Imprinting requires that the testis secrete an adequate amount of testosterone, that target organs be exposed to sufficient quantities of testosterone or testosterone metabolites, and that responsiveness of these organs to steroids not be impaired (Wilson et al. 1981). Many mechanisms may be involved in the endocrine disrupting of sex hormones. The chemical may interfere by mimicking the effects of natural hormones, binding to the hormone receptor, antagonizing the effects of natural hormones by blocking the binding to the hormone receptor, reacting directly/indirectly with the hormone structure to alter it, interfering with hormone synthesis, altering hormone receptor levels or by interfering with the transport and elimination of hormones (Baker 2001). The male reproduction system would be the target of the lesion produced by an endocrine disruptor during the hemato-testis barrier crossing, directly on the Leydig cells (do not protect by hemato-testis barrier), through the cell division or differentiation inhibition, of the metabolism intermediate interference or on the

androgen production. All those forms may affect spermatogenesis (Wilson et al. 1981). The functional activity of the male reproductive tract structures is under testosterone control. Spermatogenesis in seminiferous tubules, sperm maturation in the epididymis and the secretory activity of the accessory sex glands require adequate levels of testosterone (Waller et al. 1995).

Reproductive dysfunction is typically characterized by alterations in serum steroid hormone levels, disruptions in spermatogenesis and loss of fertility. In this work the mechanisms involved in these reproductive outcomes remain unclear. However, because testicular Leydig cells play a crucial role by producing testosterone, the mouse MA-10 Leydig tumor cell line has been used to study the molecular events involved in pesticide-induced alterations in steroid hormone biosynthesis (Walsh et al. 2000). In vivo and in vitro studies are needed to clarify the endocrine disruption produced by pesticides (Gray 1998). The Roundup® formulation (180 g/l glyphosate) showed the capacity to inhibit steroidogenesis by disrupting StAR protein expression in an in vitro assay. Although Roundup® decreased steroidogenesis, the active ingredient of this herbicide, glyphosate, did not alter steroid production, indicating that at least one other component of the formulation is required to disrupt steroidogenesis (Walsh et al. 2000). Richard et al. (2005), showed that glyphosate acts as a disruptor of mammalian cytochrome P450 aromatase activity from concentrations 100 times lower than the recommended use in agriculture on human placental cells in vitro assay. Its effects are allowed and amplified by adjuvants present in Roundup®. During the brain sex-differentiation, high concentrations of the testosterone are converted in dihydrotestosterone by the 5- α -reductase enzyme in the hypophysis and the brain stem, or they are also converted in estradiol by the aromatase enzyme in the hypothalamic and limbic system neurons. The estradiol is an important way to masculinization of the brain during this critical period. Substances capable of decreasing these enzymes and consequently the conversion in dihydrotestosterone and estradiol, inhibit the sex differentiation for the male (Kelly 1991).

Despite the fact that the doses used in this study would never be expected to correspond to human exposure levels under normal circumstances, as reported by Williams et al. (2000) for glyphosate and polyoxyethyleneamine in adults or children (margins of exposure = 5,420, 3,370 and 461, 577, respectively), the results by in vivo study shows that the commercial formulation poses an increased potential risk for the rat reproductive system; likewise are the results by in vitro assay showed by Walsh et al. (2000) and Richard et al. (2005). To better understand this fact, additional studies should be carried out to determine the mechanisms of effects of the commercial formulation on the reproductive system of the male exposed in utero and at lactation period, since humans are also potentially exposed to this commercial formulation.

Conclusions

The results of this study showed that pre- and postnatal exposure to the herbicide glyphosate Roundup® can induce adverse effects on the reproductive system of male offspring without inducing maternal toxicity. To female offspring only the vaginal canal opening was delayed.

The doses of glyphosate Roundup® used in this in vivo study (50, 150 and 450 mg/kg) produced a decrease in sperm number per epididymis tail and in daily sperm production during adulthood in the exposed rat offspring, plus a dose-related decrease in the testosterone serum level, an increase in the percentage of abnormal sperms at puberty and signs of spermatid degenerating in both periods. These results characterize this herbicide as a probable endocrine disruptor. However, other studies are needed to elucidate the mechanism, the main component of the commercial formulation and/or the interactions between components that exerted reproductive effects on males exposed to Roundup® during the pre- and perinatal periods.

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